

Marcus Brew Ashmond

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SUMMARY

Full stack and backend engineer with 3+ years of experience building and scaling distributed systems in production. Shipped microservices handling millions of daily transactions, reduced API latency by 40%, and led engineering work across fintech, healthtech, and AI projects. Comfortable across the full stack, from designing REST APIs and event driven architectures in Go and Python to building React frontends and deploying containerized services on Kubernetes.

EXPERIENCE

Software Engineer

Sep 2023 – Aug 2025

Betika (Leading African fintech platform, 5M+ users)

Remote, Kenya

- Designed and deployed high performance **Go** microservices handling millions of daily transactions, improving system throughput by **35%** through async processing and optimized service boundaries.
- Overhauled **MySQL** indexing, query plans, and connection pooling, cutting API latency by **40%** during peak traffic for **100,000+ concurrent users**.
- Engineered resilient **Kafka** consumer patterns with retry logic and dead letter queues, reducing message failures by **90%**.
- Containerized all services with **Docker** and orchestrated on **Kubernetes**, cutting release cycles by **60%** through **CI/CD** pipelines that enabled zero downtime rollouts.
- Led code reviews and established an automated testing culture across the engineering team, dropping production defects by **30%**.

Software Engineering Intern

Dec 2023 – Jan 2024

Cognizant

Teaneck, NJ

- Built **Flask** APIs integrated with **Azure** OpenAI for automated document parsing and summarization, cutting analysis time by **65%**.
- Engineered a **React** and **TypeScript** frontend with optimized state management, reducing user query complexity by **33%** and improving UX clarity.
- Deployed secure, production ready cloud solutions on **Azure** with **CI/CD** pipelines alongside senior engineers on the team.

Full Stack Developer

Aug 2022 – Aug 2023

Vien Health

Remote, Ghana

- Built a **HIPAA** aware telemedicine platform supporting **1,000+ monthly** video consultations, integrating **Twilio** WebRTC for real time video with latency under 300ms.
- Designed secure **RESTful APIs** for scheduling, patient authentication, and EMR using **Node.js** and **MySQL**, improving clinical workflow efficiency by **35%**.
- Implemented **PCI** compliant payment processing with **Stripe** for **\$100,000+** in annual transactions, including full audit logging and end to end encryption.
- Built automated testing and monitoring in **React** and **TypeScript** that increased platform reliability.

EDUCATION

Bachelor of Science in Computer Science — GPA: 4.0

Graduated 2024

University of Maryland Global Campus

PROJECTS

FinSight — Personal Finance Dashboard

React, Node.js, MongoDB, JWT | [Live Demo](#) | [GitHub](#)

- Built and deployed a full stack personal finance platform covering budget tracking, transaction management, net worth goals, CSV upload, dark mode, and mobile responsive design using **React**, **Node.js**, and **MongoDB** Atlas.
- Implemented **JWT** authentication and shipped through 6 commits from MVP to production, deployed live on **Vercel**.

FHIR Health Data Interoperability Service

Go, PostgreSQL, Kafka, Docker, Kubernetes | [GitHub](#)

- Built a **FHIR** compliant **REST API** in **Go** supporting full CRUD on patient and clinical observation data, backed by **PostgreSQL** persistence.
- Engineered a real time **Kafka** event pipeline that publishes on observation creation and a consumer service that flags abnormal vital signs.
- Implemented **HIPAA** aligned audit logging middleware recording every access to protected health data, and containerized the service with a multistage **Docker** build deployed via **Kubernetes** manifests, health probes, and resource limits, backed by automated tests and CI.

TECHNICAL SKILLS

Languages: Go, Python, JavaScript, TypeScript, Java, C/C++, SQL

Backend and Architecture: Microservices, REST APIs, gRPC, Event Driven Systems, Distributed Systems, System Design

Frameworks and Tools: React, Next.js, Node.js, Express.js, Flask, Kafka, RabbitMQ, Docker, Kubernetes

Databases: MySQL, MongoDB, Redis, PostgreSQL, DynamoDB

Cloud and DevOps: AWS, Azure, GCP, Terraform, CI/CD, Linux, Vercel

CERTIFICATIONS

Microsoft Azure Security Engineer (AZ 500) | CompTIA Security+ | CompTIA Network+